

CCS Theia

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CCS Theia (Code Composer Studio IDE Theia): An integrated development environment (IDE) for TI microcontrollers and processors.

MSPM0 Software Development Kit (SDK): Provides comprehensive software resources, along with corresponding tools and documentation.

The installation package path and software installation path should not contain Chinese characters. Please place the installation package in a fully English path for installation.

1. Download MSPM0-SDK

Download address: [MSPM0-SDK Software development kit \(SDK\) | TI.com](#)

You can download the latest version from the above link, or use the version provided in our materials.



Get started

- Step 1: Get your LaunchPad™ development kit
- Step 2: Download the MSPM0 SDK or [browse the SDK online](#)
- Step 3: Evaluate code examples using our [QuickStart guides](#)

Downloads



SOFTWARE DEVELOPMENT KIT (SDK)

MSPM0-SDK — MSPM0 Software Development Kit (SDK)

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MSPM0-SDK — MSPM0 Software Development Kit (SDK)

Latest version Version: 2.03.00.07 Release date: 27 Nov 2024

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Supported products & hardware

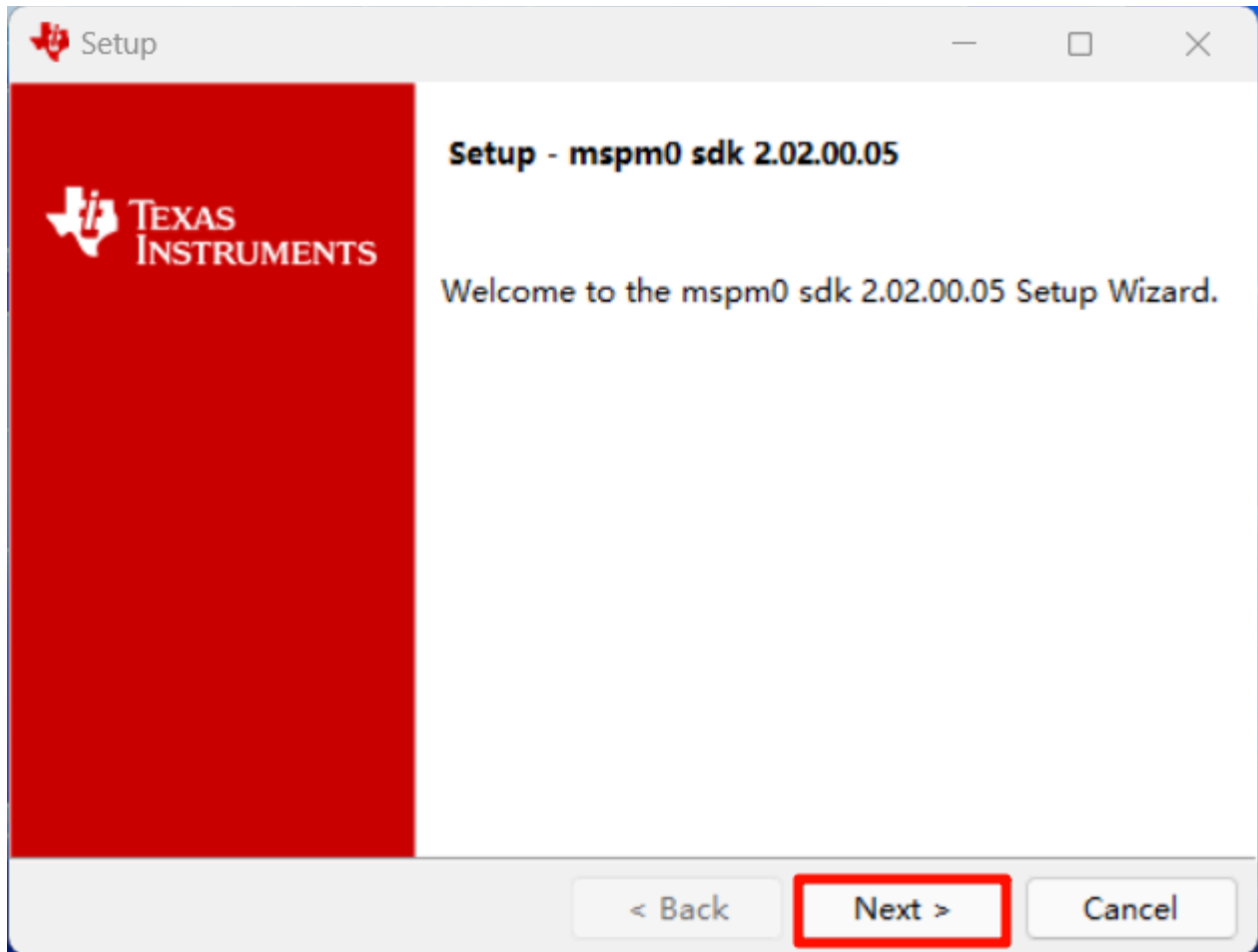
 mspm0_sdk_2_03_00_07.exe — 150560 K 	MSPM0 SDK for Windows MD5 checksum <code>d3da1eeb446534a3852a6910b5c4b097</code> 
 mspm0_sdk_2_03_00_07.run — 150734 K	MSPM0 SDK for Linux MD5 checksum <code>fa646f1eb5badae238b843e6740f1100</code> 
 mspm0_sdk_2_03_00_07.app.zip — 153171 K	MSPM0 SDK for macOS MD5 checksum <code>3fa0a67ffacb11800576b30ace795925</code> 

 = Requires export approval (1 minute)

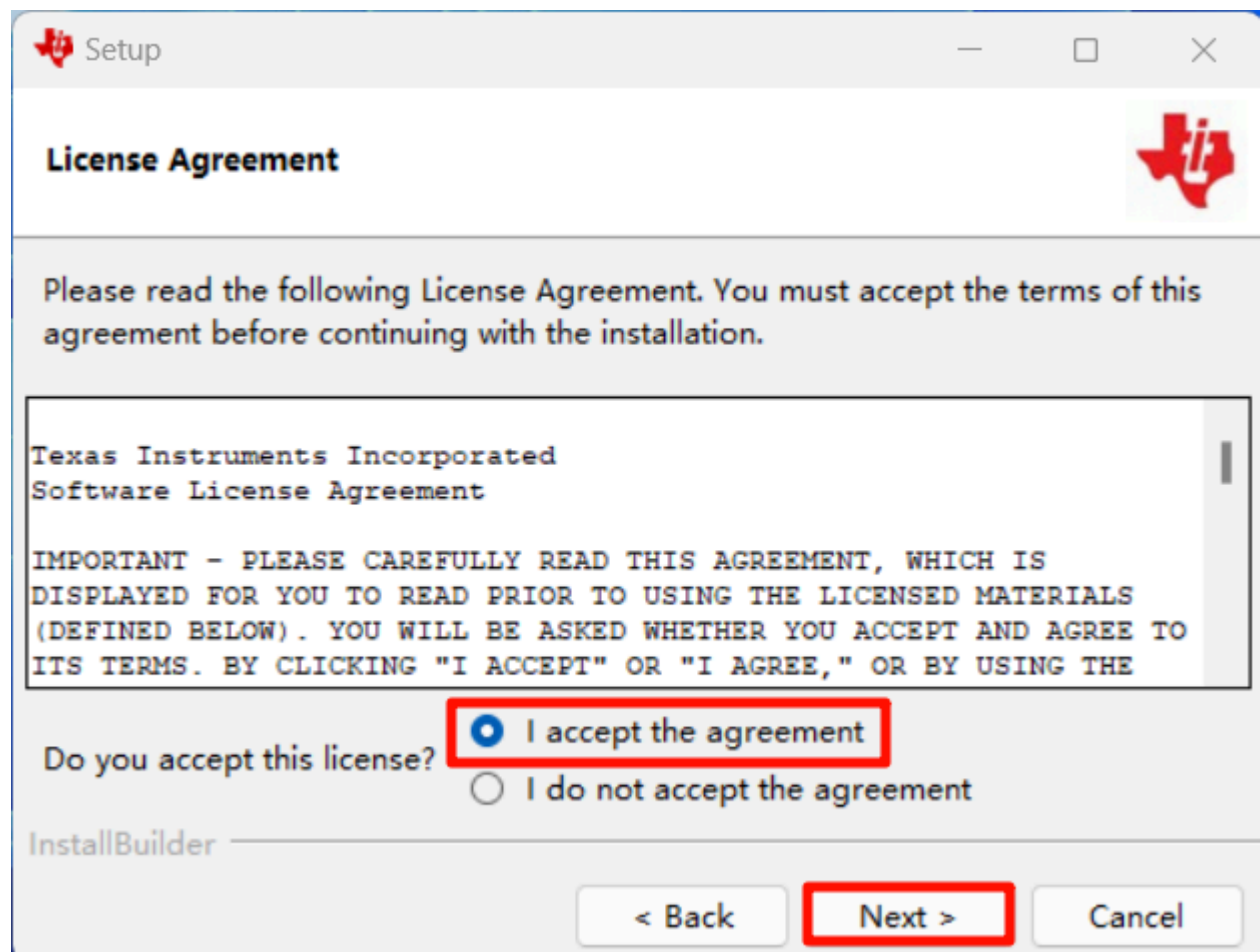
Documentation

2. Install MSPM0-SDK

Open `mspm0_sdk_xx.exe` as administrator and follow the prompts:

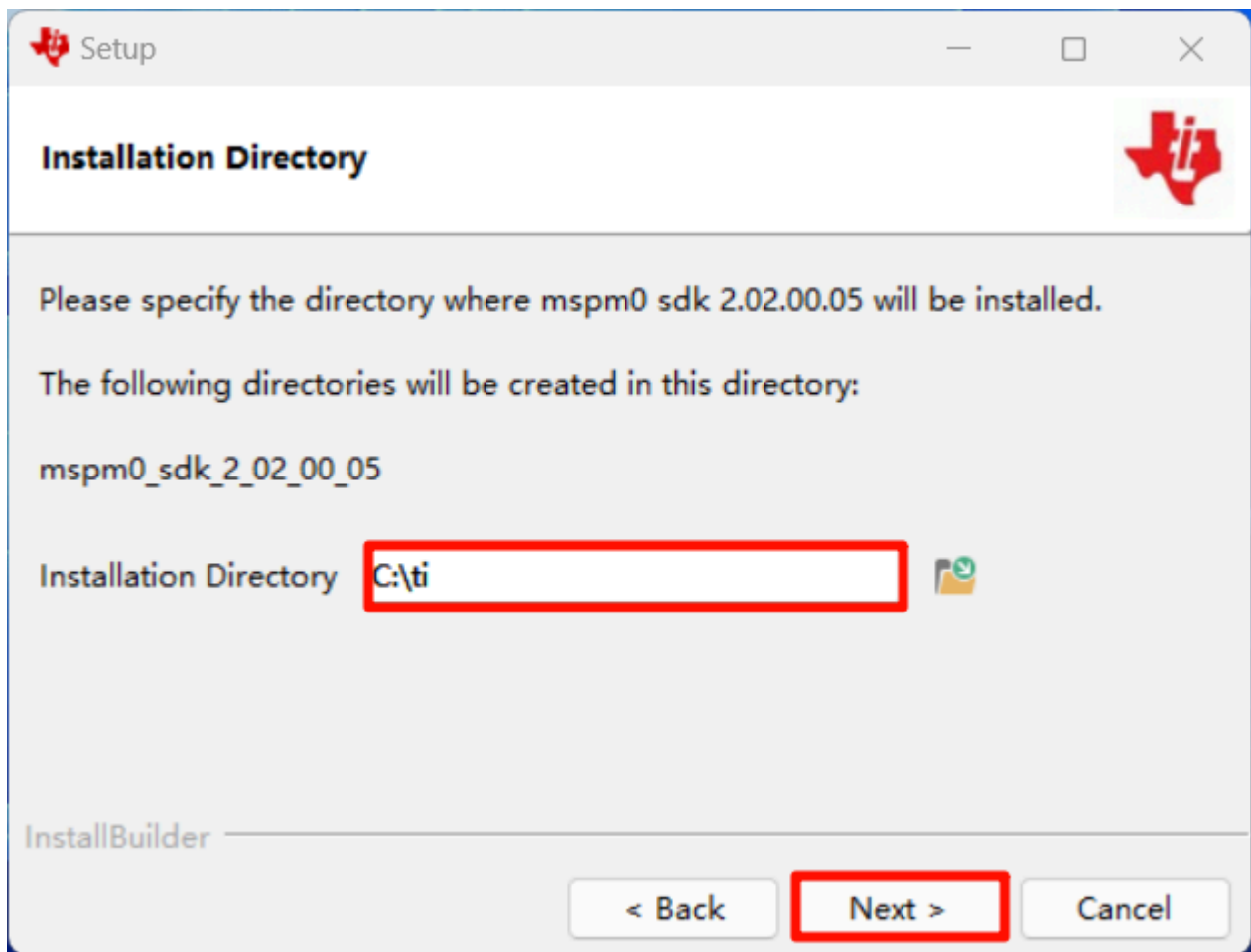


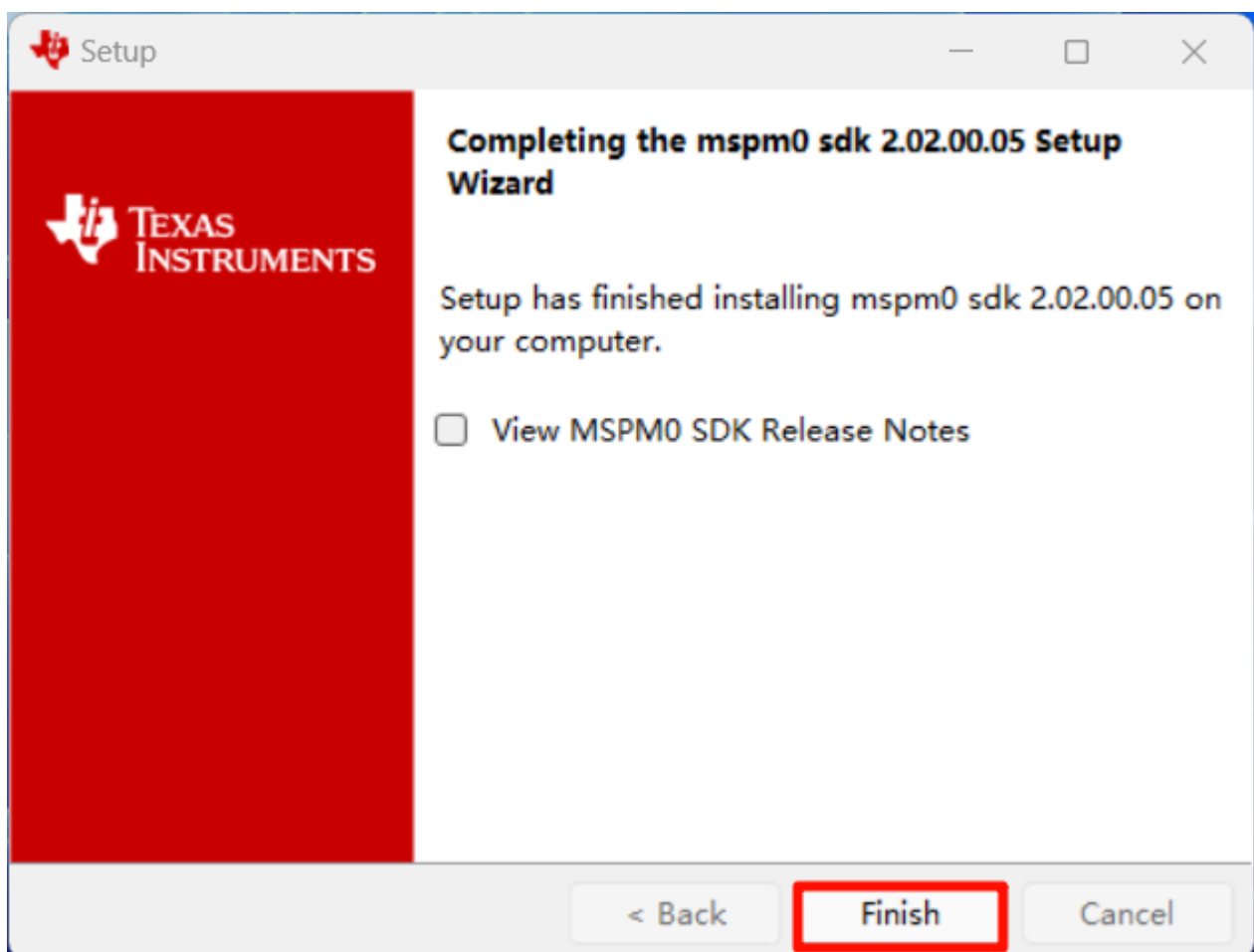
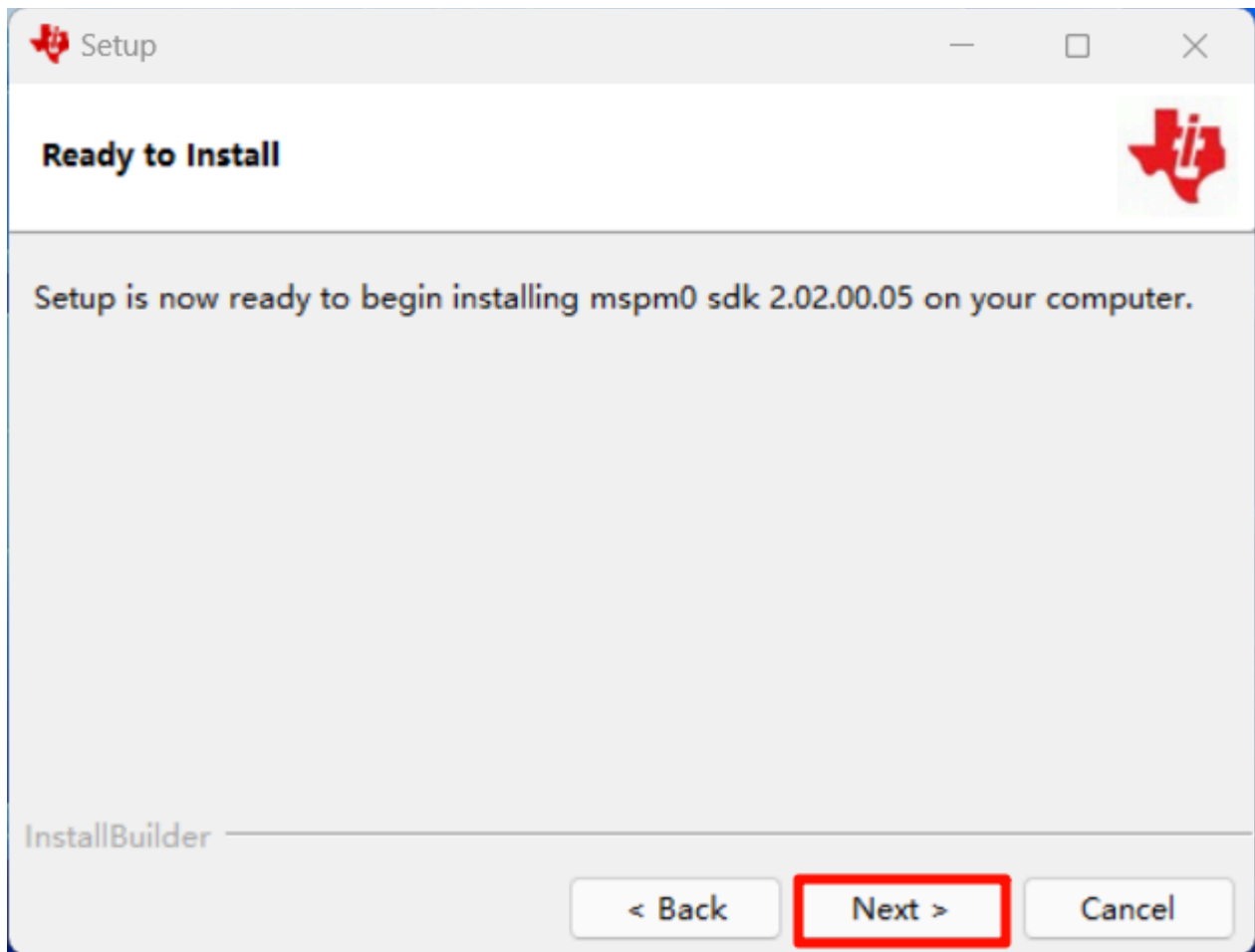
Accept Agreement



Installation Location

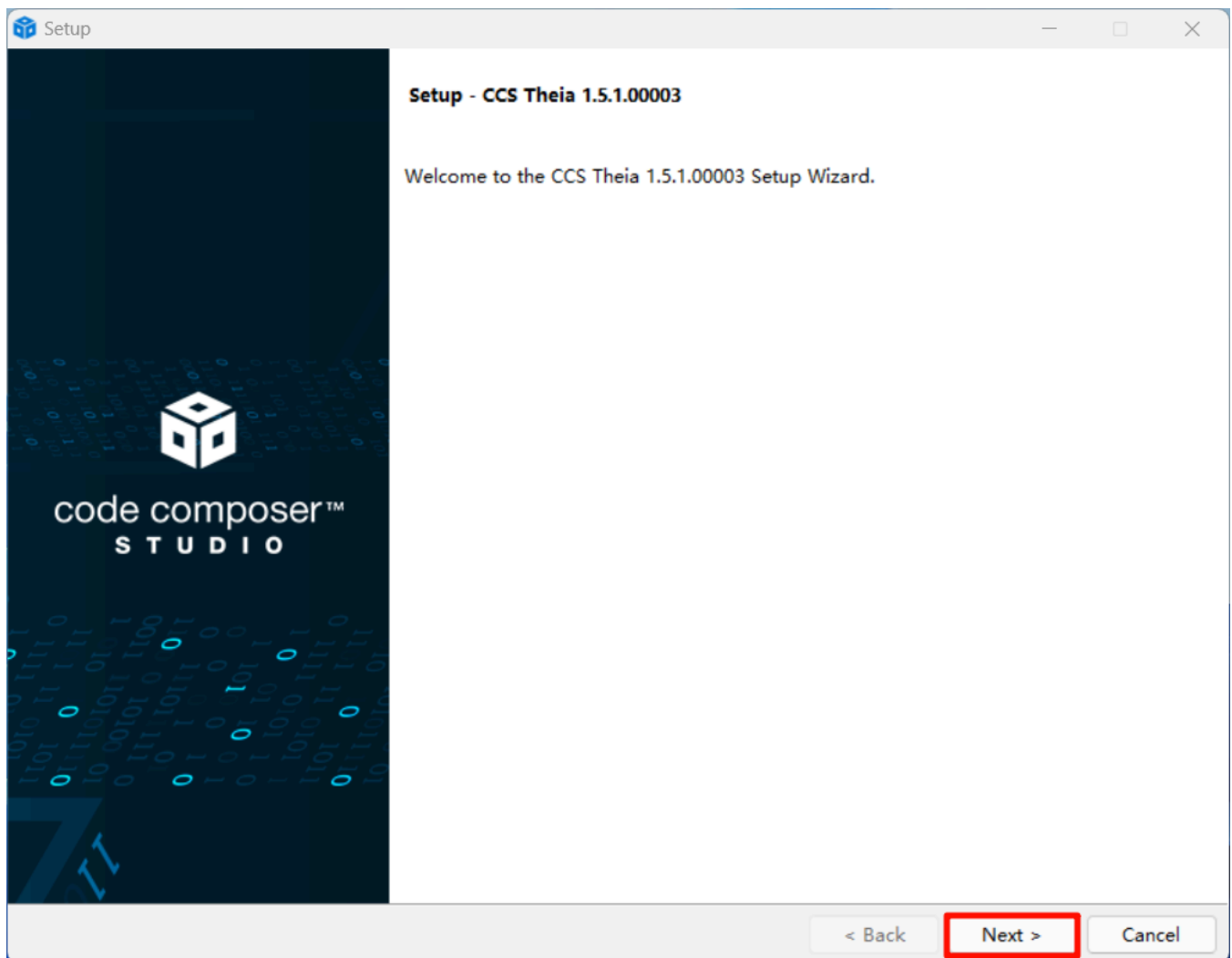
It is recommended to use the default location:





3. Install CCS Theia

Open `ccs_theia_setup_xx.exe` as administrator and follow the prompts:



Accept Agreement



License Agreement

Please read the following License Agreement. You must accept the terms of this agreement before continuing with the installation.

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Limited License Agreement.

This Limited License Agreement ("Agreement") is a legal agreement between you (either an individual or entity) and Texas Instruments Incorporated ("TI"). The "Software" consists of the following materials: (a) the materials identified as TI proprietary software programs in the software manifest for the software subject to the terms herein, and any "on-line" or electronic documentation associated with these programs, or any portion thereof (the "Licensed Materials"), and (b) the materials identified as open source materials or third party proprietary software in the software manifest for the Software, or any portion thereof ("Public Software"). For clarification, your use of the Licensed Materials is subject to the licensing terms contained in this Agreement and your use of the Public Software is subject to the separate licensing terms specified in the applicable software manifest and/or identified or included with the materials to which they apply. This Agreement does not limit your rights under, or grant you rights that supersede, the license

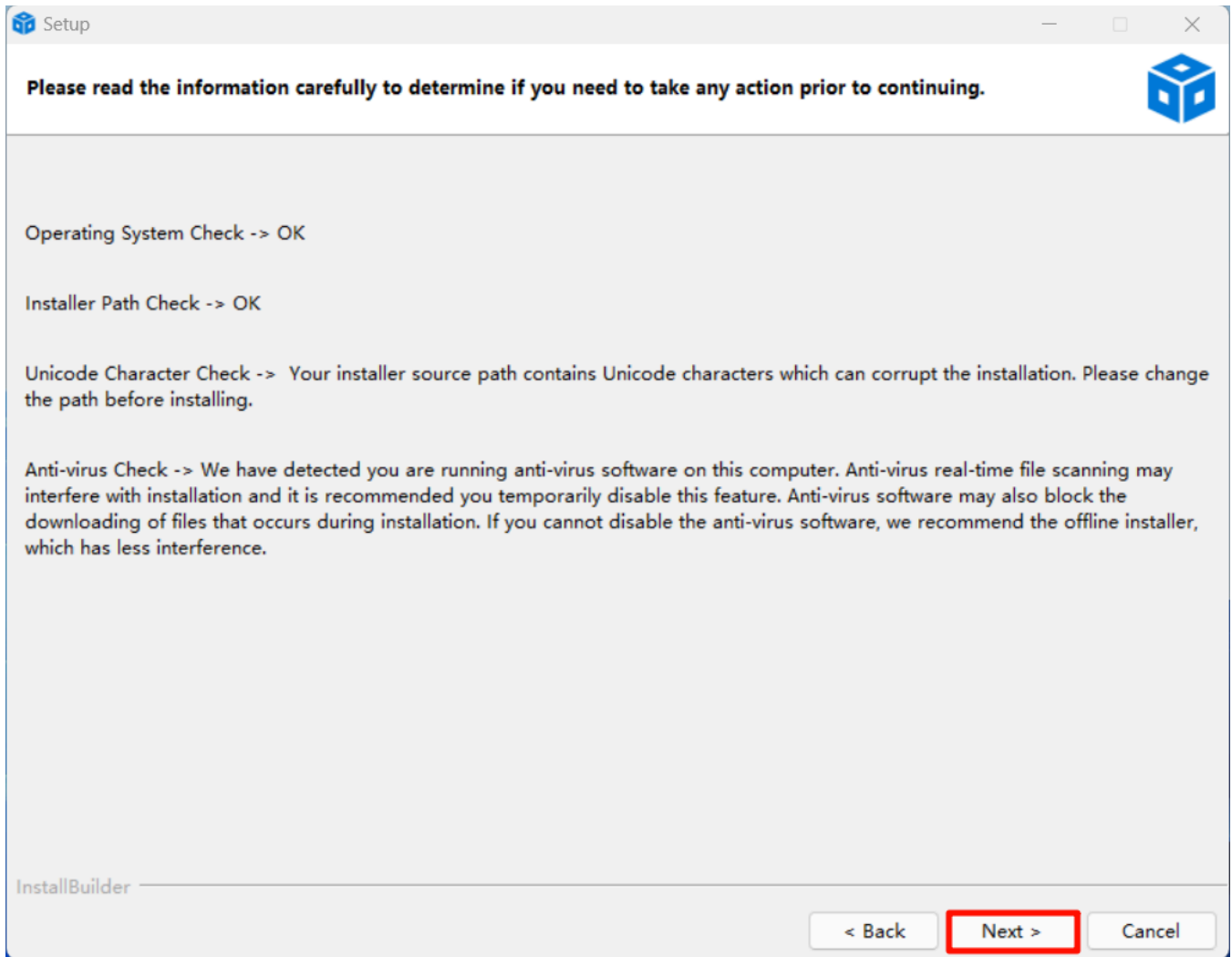
Do you accept this license? ☒ I accept the agreement
☐ I do not accept the agreement

InstallBuilder

< Back

Next >

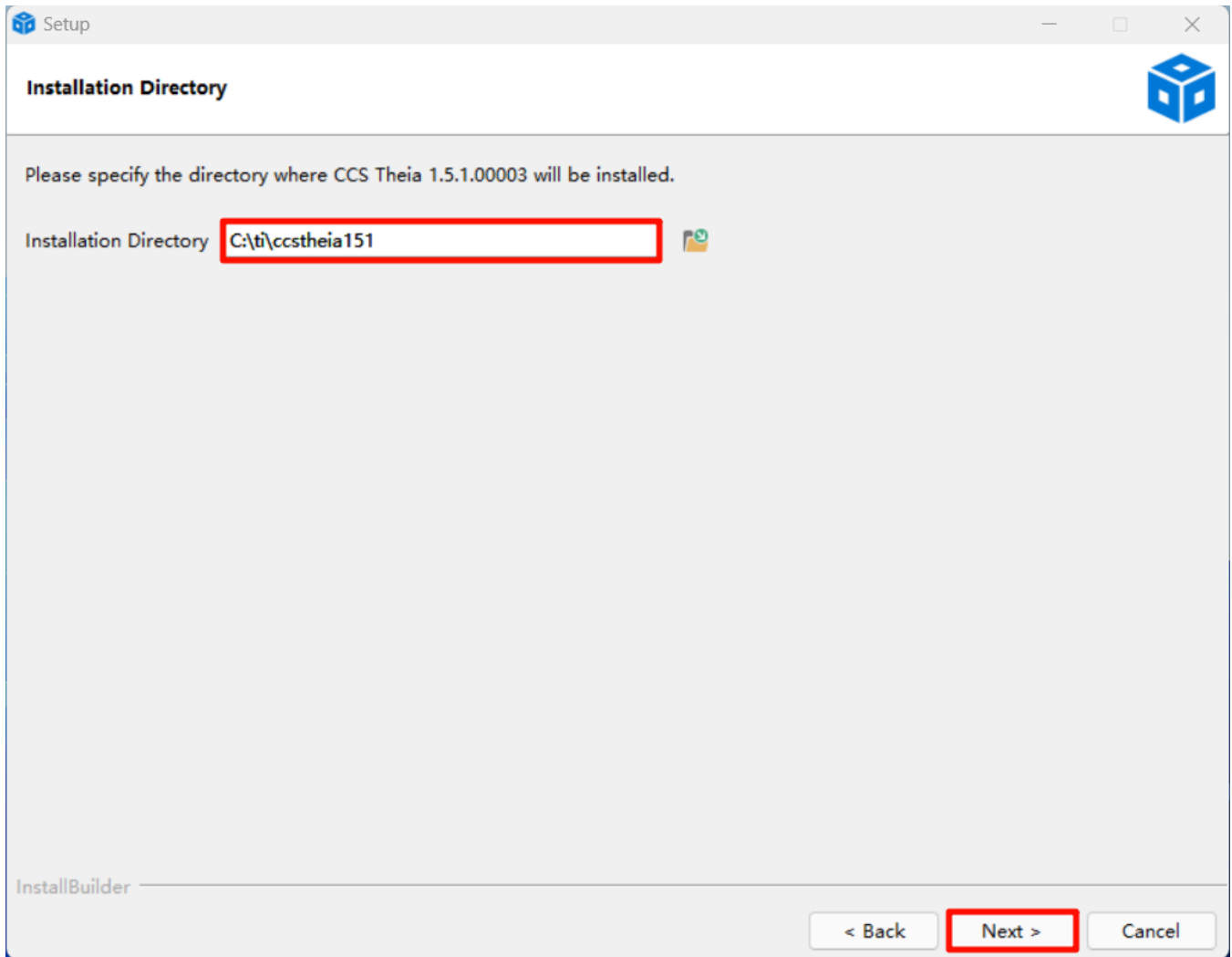
Cancel



Installation Location

It is recommended to use the default location:

Note: The path cannot contain Chinese characters. If your computer username is in Chinese, you cannot install in the default location. You must install in an English path.



Install Components

**Select Components**

Select the components you want to install; clear the components you do not want to install. Click Next when you are ready to continue.

- ☒ MSPM0 Arm® Cortex®-M0+ microcontrollers
- ☐ MSP430™ microcontrollers
- ☐ C2000™ real-time microcontrollers
- ☐ Wireless Connectivity
- ☐ Arm®-based processors
- ☐ AM2x Arm®-based high performance microcontrollers
- ☐ mmWave Sensors

Click on a component to get a detailed description

**Ready to Install**

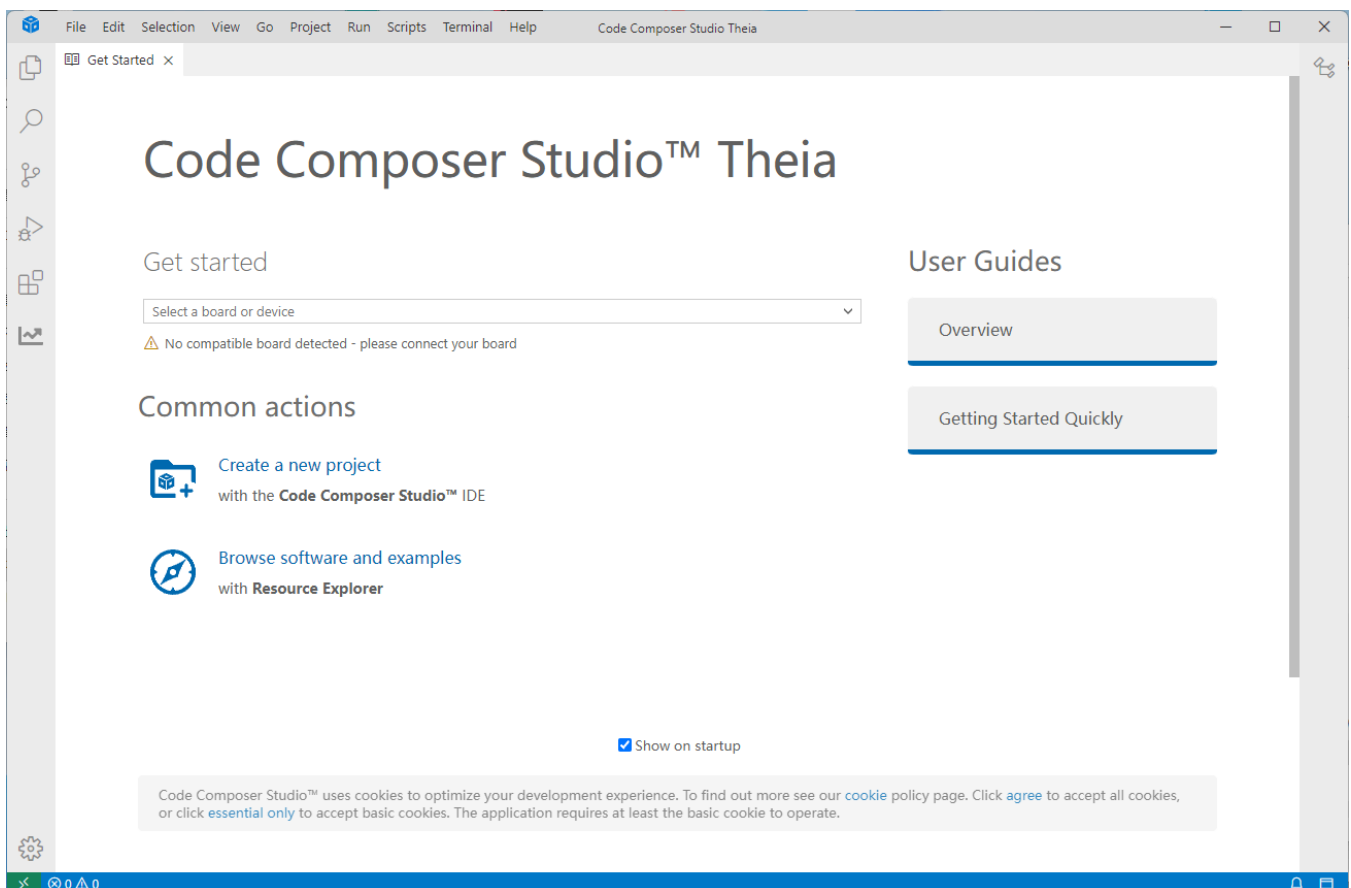
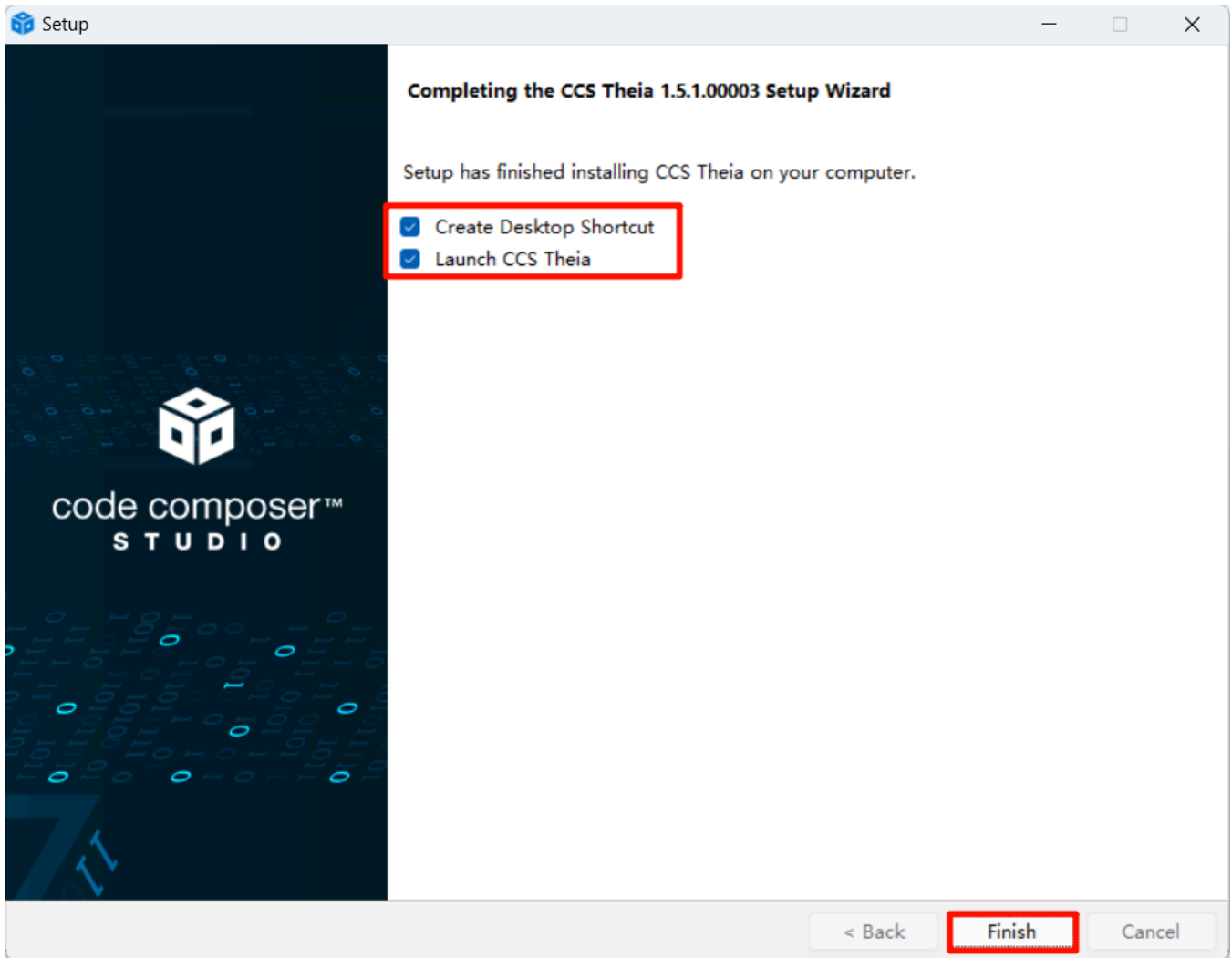
Setup is now ready to begin installing CCS Theia 1.5.1.00003 on your computer.

InstallBuilder

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Next >

Cancel

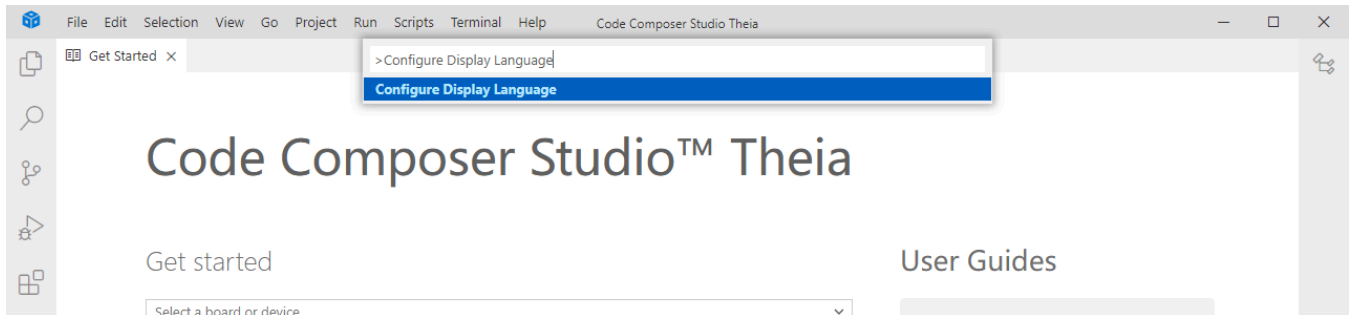


4. Manual SDK Configuration

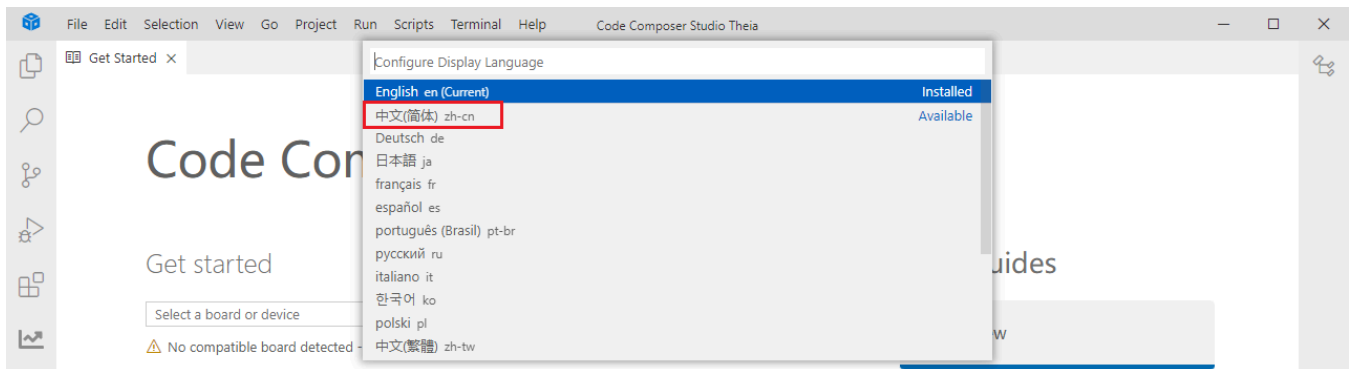
4.1 Configure other language Display*

You can skip this configuration tutorial. It does not affect the environment setup.

Press the shortcut key shift+ctrl+P, enter `Configure Display Language` to access the language configuration interface.

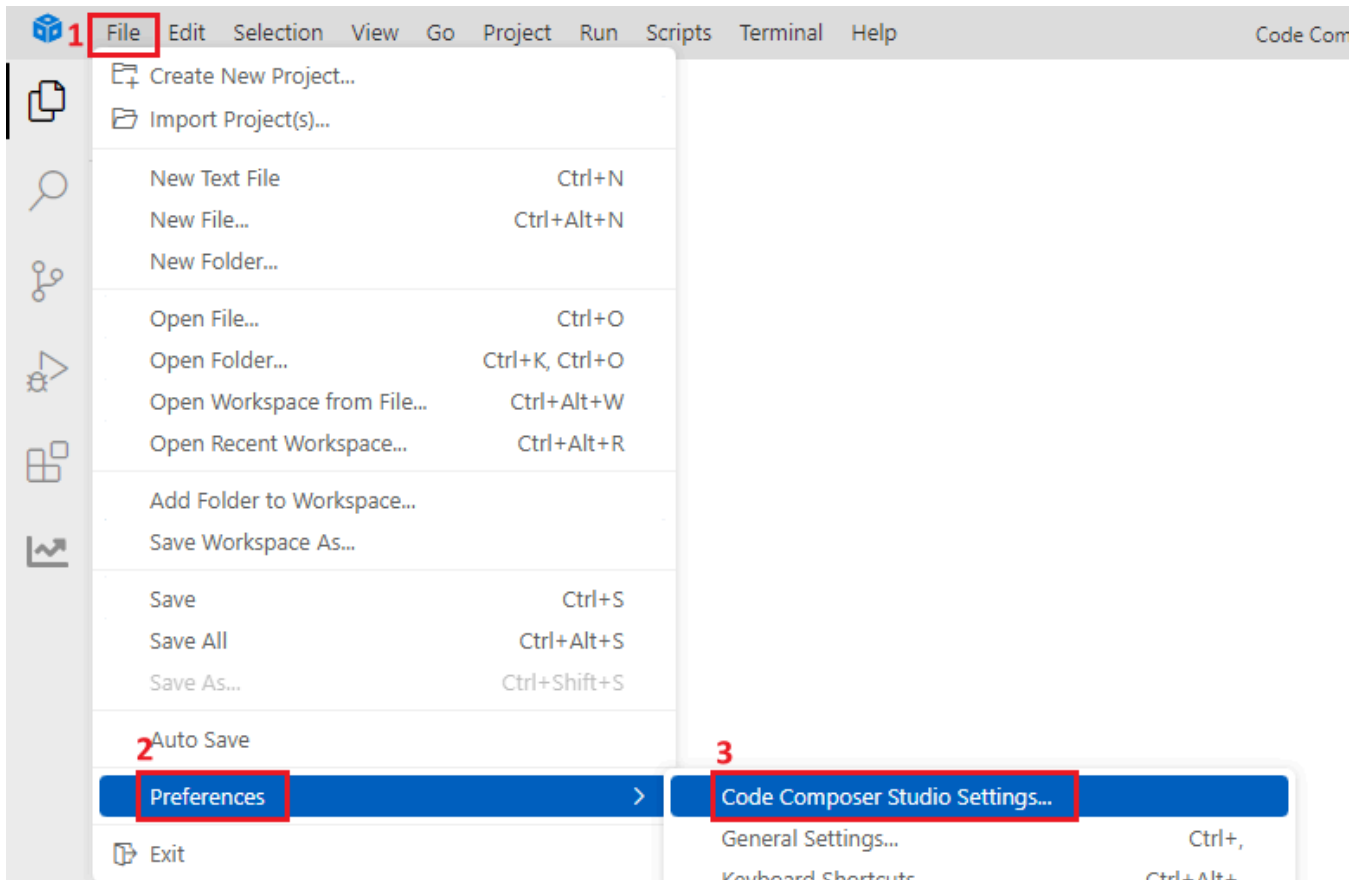


Wait for CCS to load all languages, then select Chinese to install.

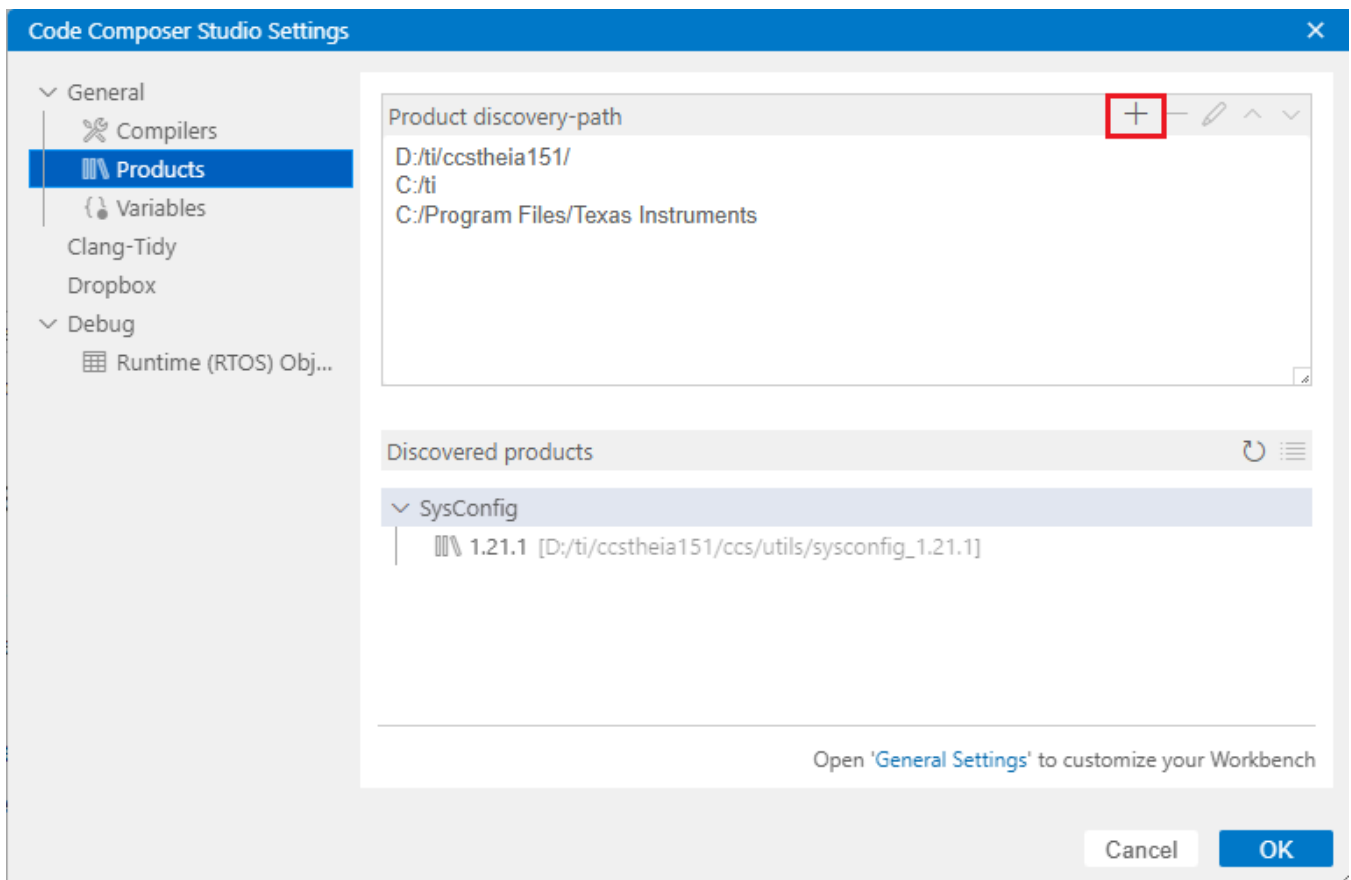


4.2 Configure SDK

Open CCS Preferences --> Code Composer Studio Settings...

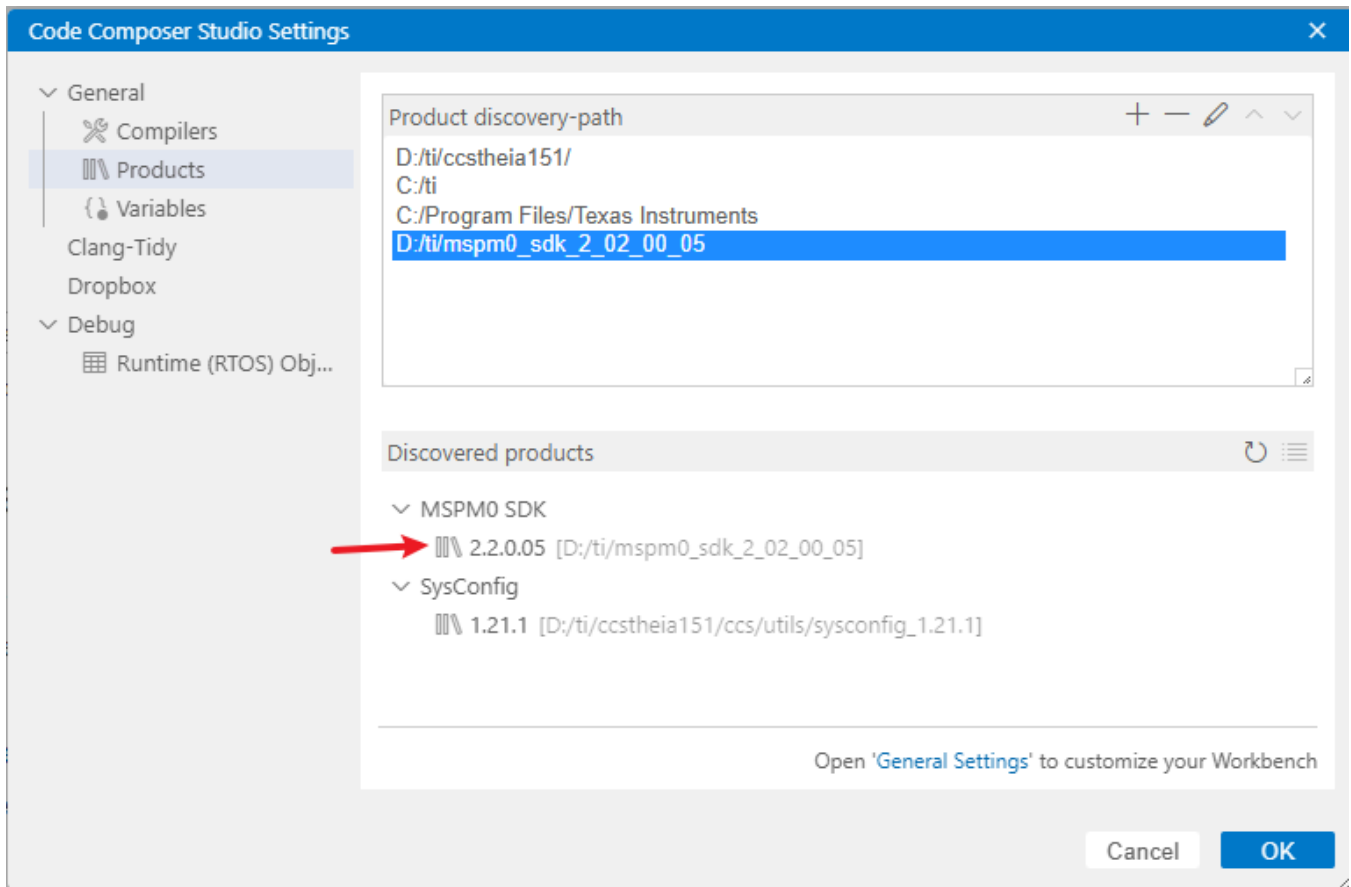


Click **Products**, add the SDK path.



Select your SDK path and import it into CCS configuration. The SDK will be automatically recognized.

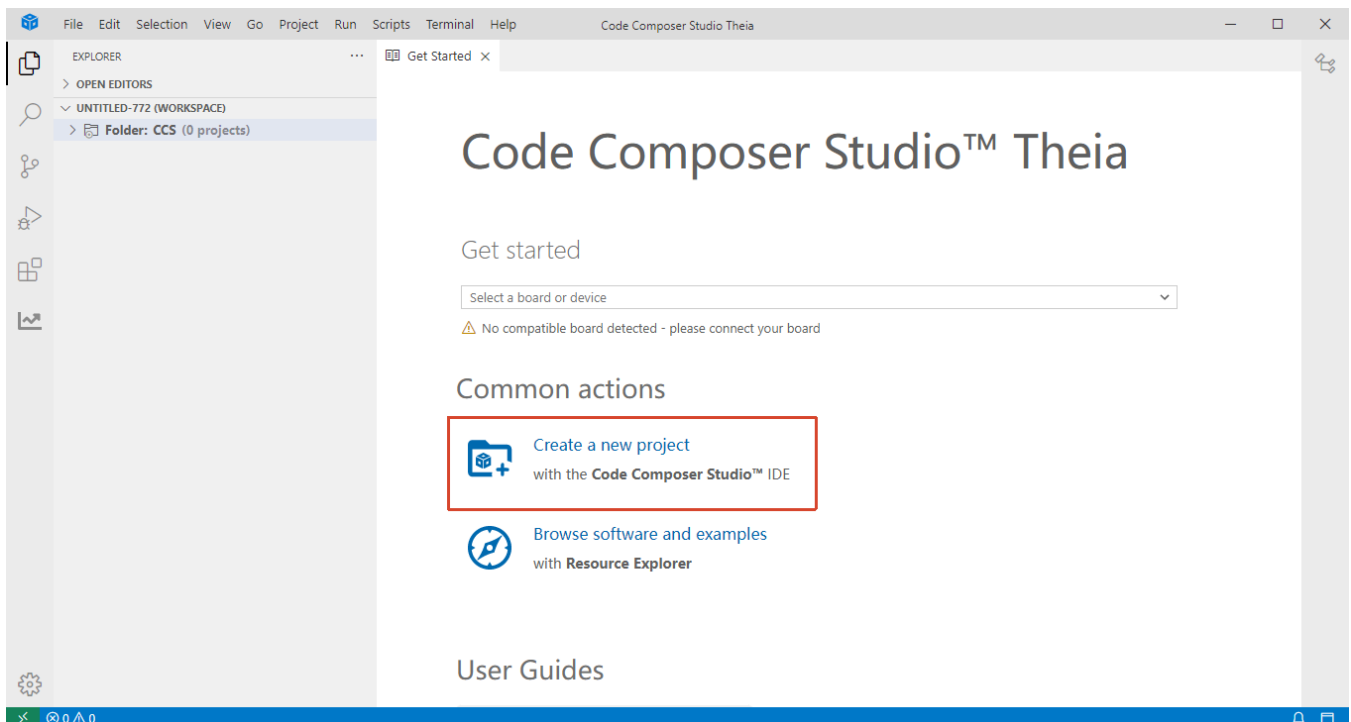
The following shows my SDK path address.



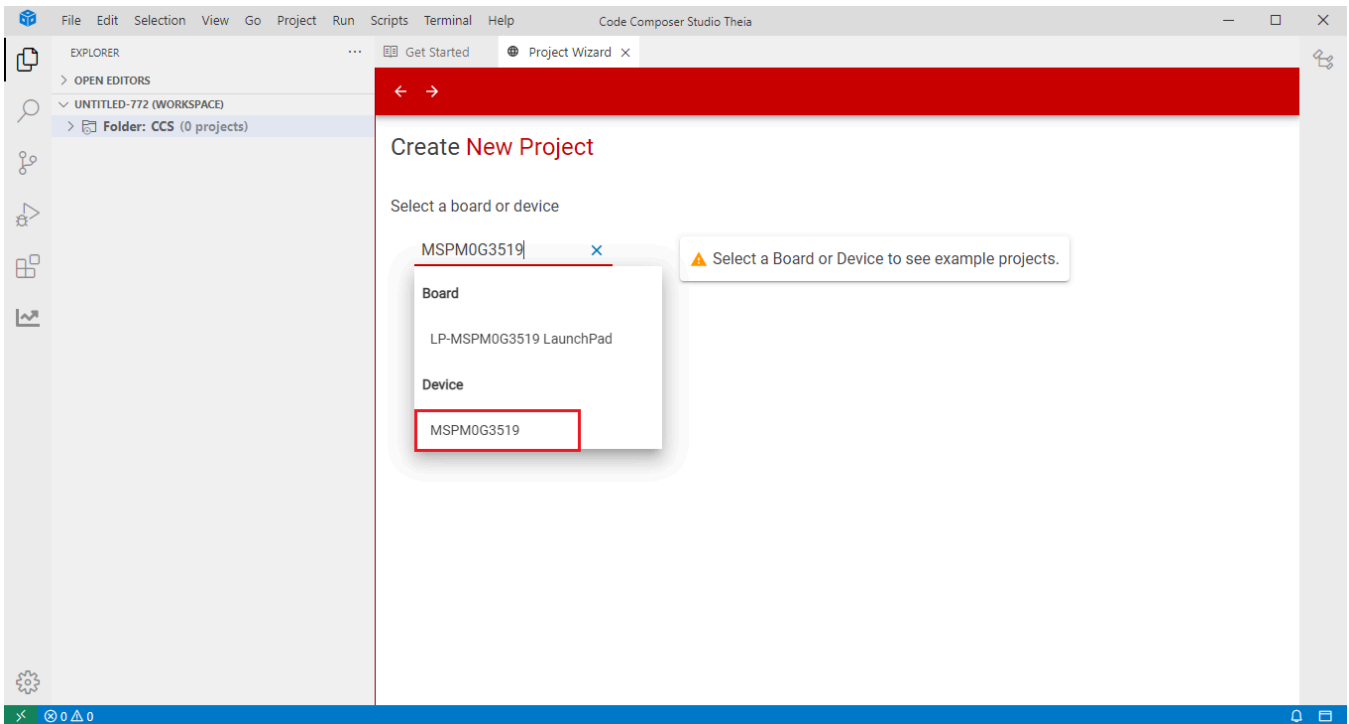
After completion, click OK.

5. Create New Project

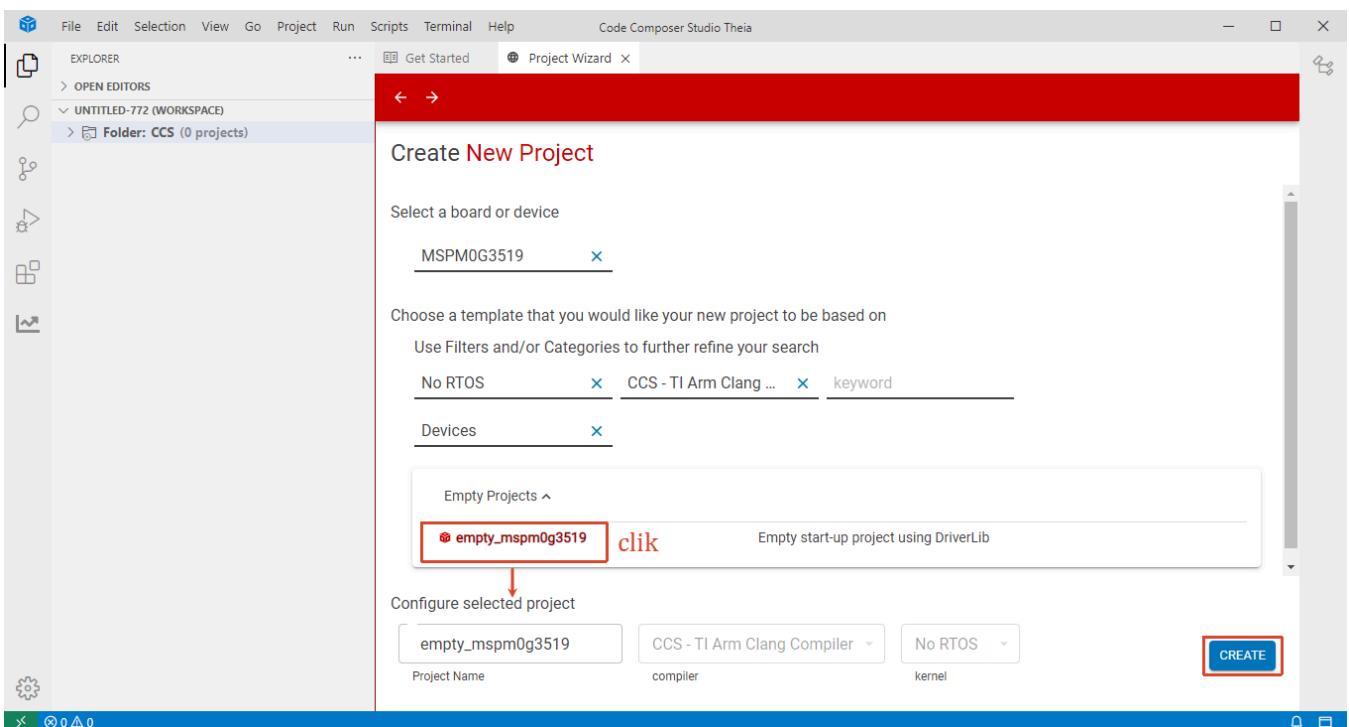
Click **Create a new project** to create a new project



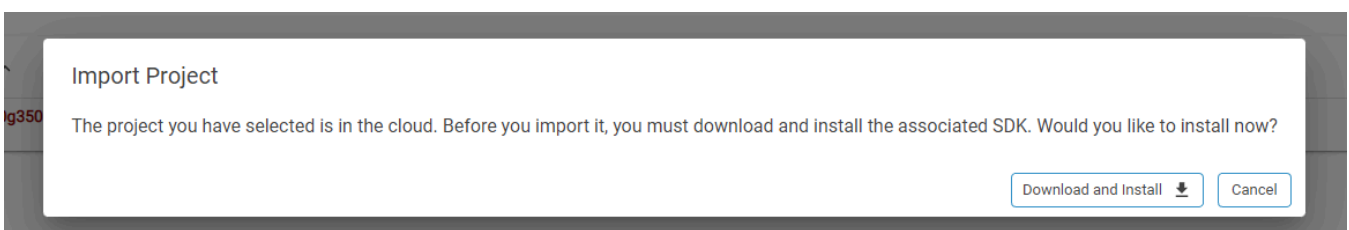
Manually enter MSPM0G3519, and select MSPM0G3519 as the device from the dropdown.



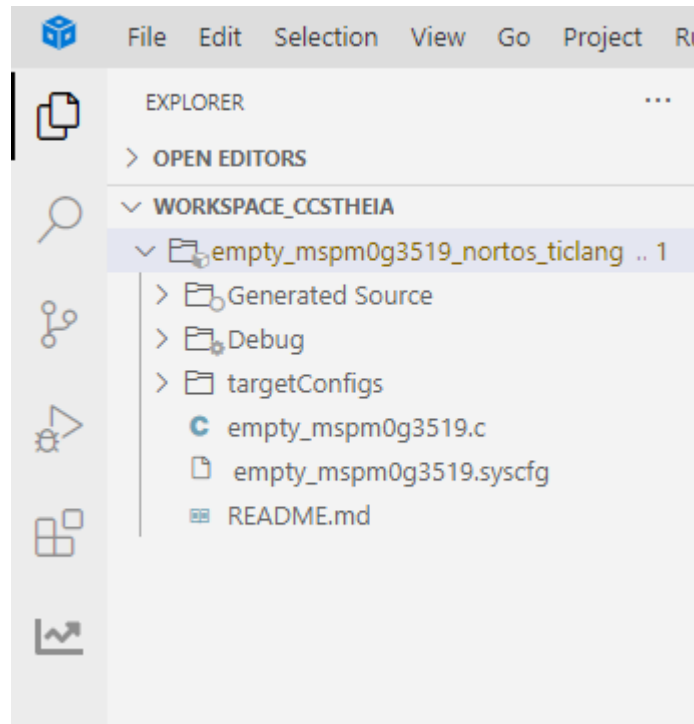
Then set the parameters as follows. After setting, click **CREATE** to create the project.



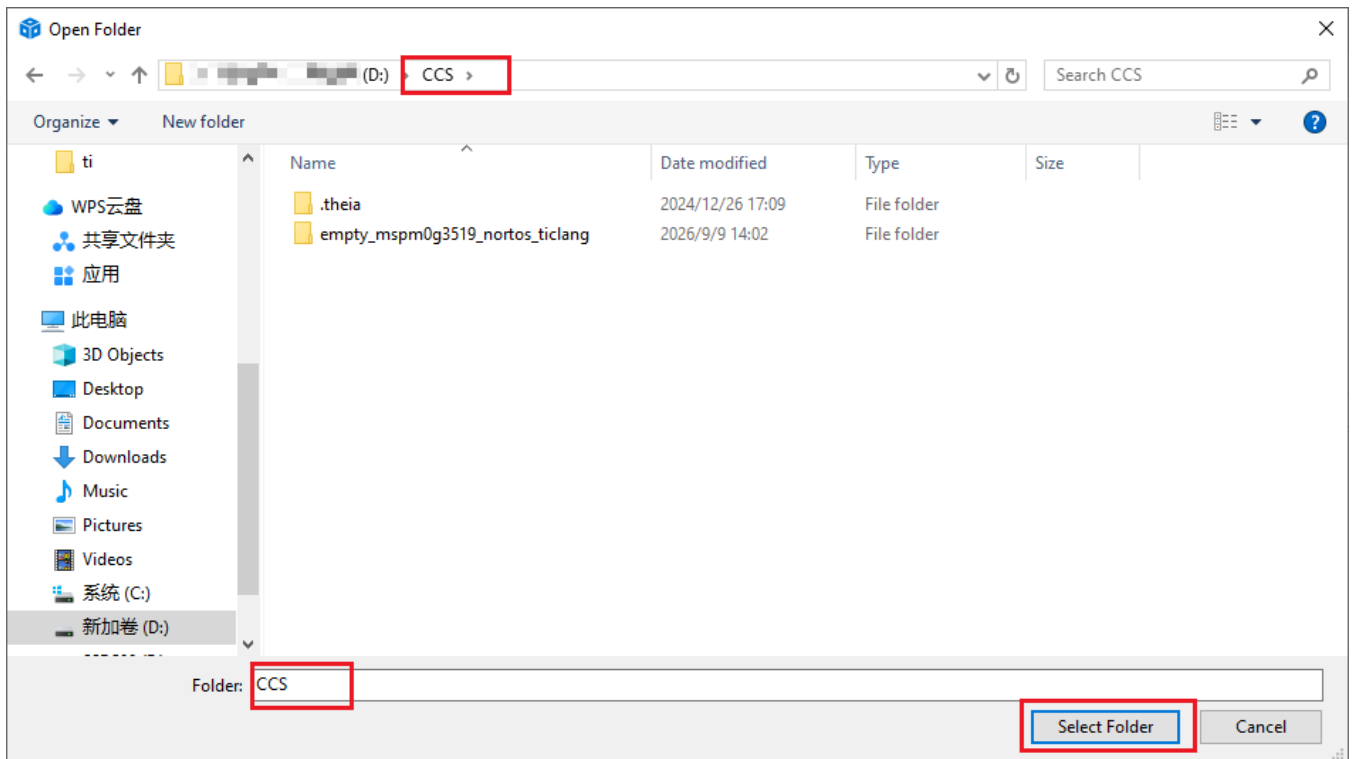
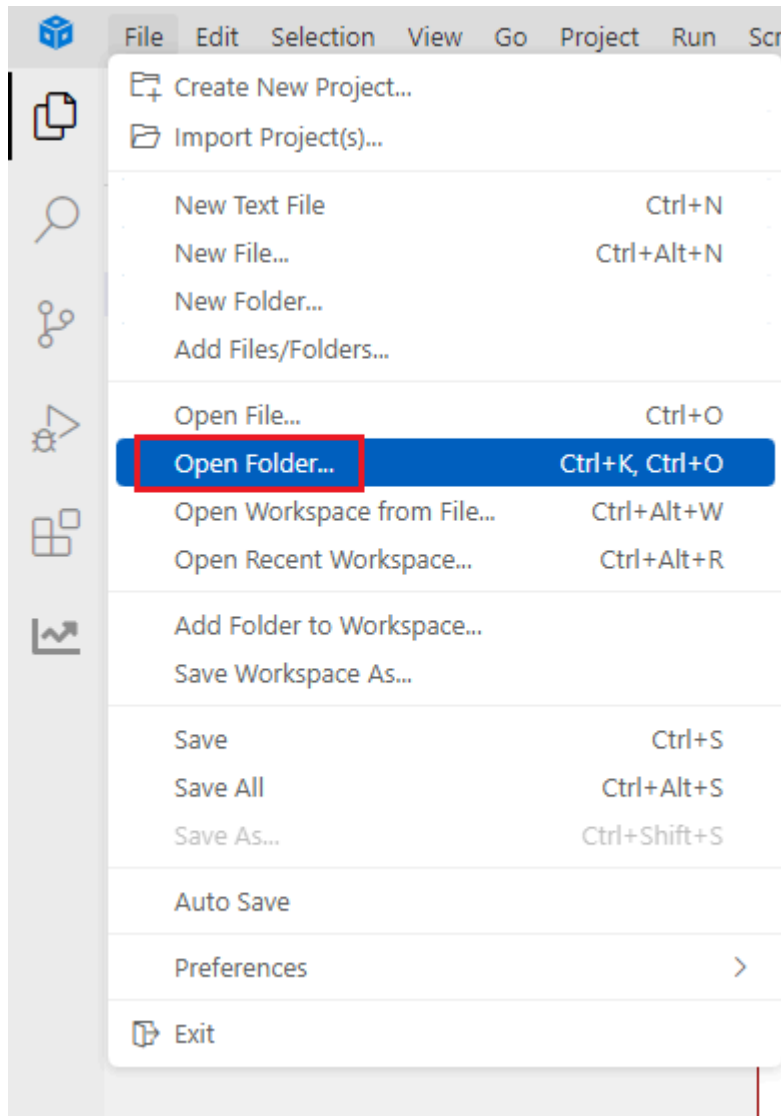
During creation, it may prompt that no SDK is detected and ask if you want to install online. Since you have manually configured the SDK earlier, if this prompt appears, it means the SDK is the latest version. You can download the latest version from the SDK download link and reconfigure.

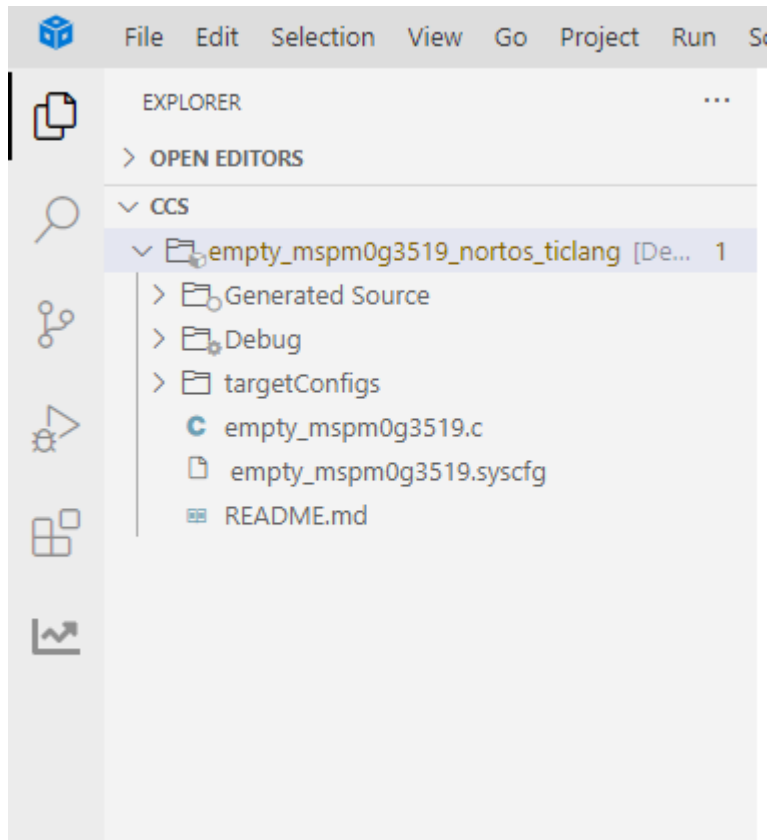


This is the project we created.



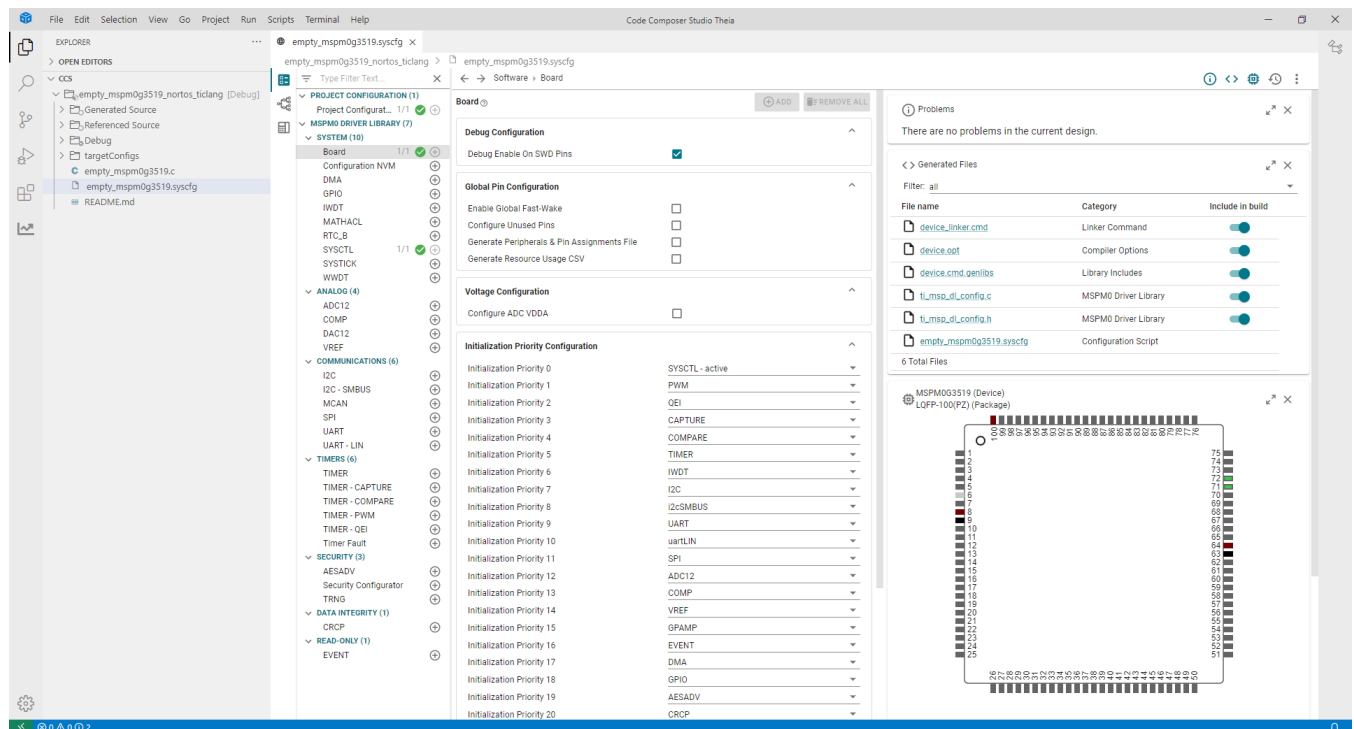
The default location for newly created projects when first opening the software is C:\Users\Administrator\workspace_ccstheia (workspace). You can copy the project file to another folder, then open that folder in CCS Theia to generate a new workspace. The next time you open the software, it will open the new workspace by default, and newly created projects will also be placed in that workspace by default.





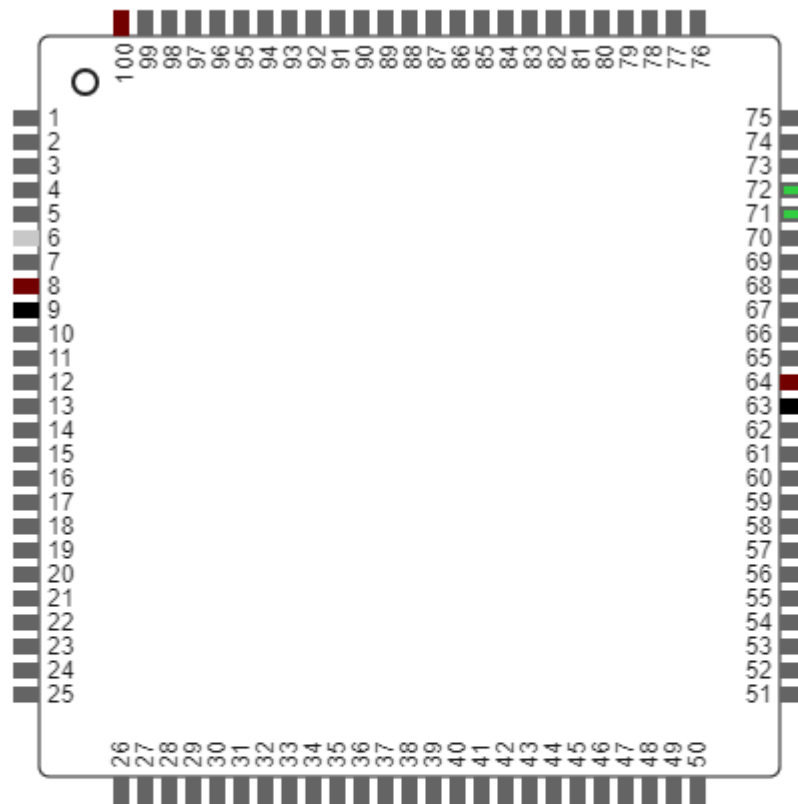
6. SYSCFG Settings

Open the .syscfg file of the project. You can configure on this page. After configuration, use the shortcut key **ctrl + s** to save your configuration.



Note: You need to select the corresponding configuration here based on the chip package you are using. The chip package used by Yahboom is **LQFP-48(PT)**.

MSPM0G3519 (Device)
LQFP-100(PZ) (Package)



SWITCH

Pin Available



Pin Assigned



Warning



Power



Ground



Fixed (N/A)



Migrate Settings

This will migrate the current configuration to the board or device selected below. Any incompatibilities will be flagged as errors.

The migration can be undone by using ctrl + z or the history view. Once satisfied with the conversion, saving the changes will initiate the migration of the underlying project which cannot be undone.

See [MSPM0 SDK CCS IDE Guide](#) for extended details on Migrating Between MSPM0 Derivatives.

Setting	Current Value	New Value
Board		None
Device	MSPM0G3519	MSPM0G3519
Package	LQFP-100(PZ)	LQFP-100(PZ)
Lock Resource Allocation		

LQFP-48(PT)

LQFP-64(PM)

LQFP-64(PTE)

LQFP-80(PN)

LQFP-100(PZ)

VQFN-32(RHB)

VQFN-48(RGZ)

ANCEL

CONFIRM

erty 9

UART

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uartLIN

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SPI

26

27

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32

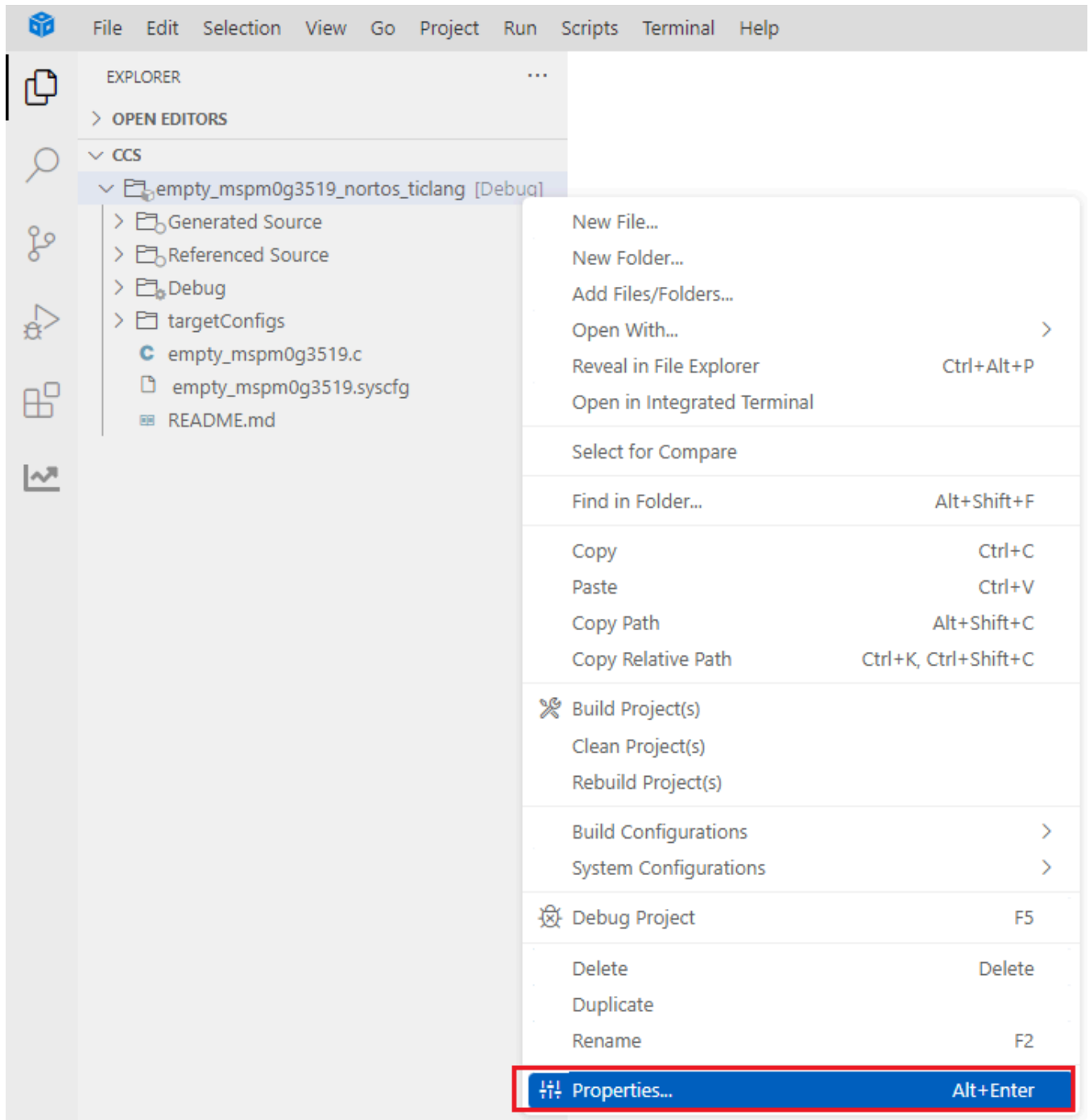
33

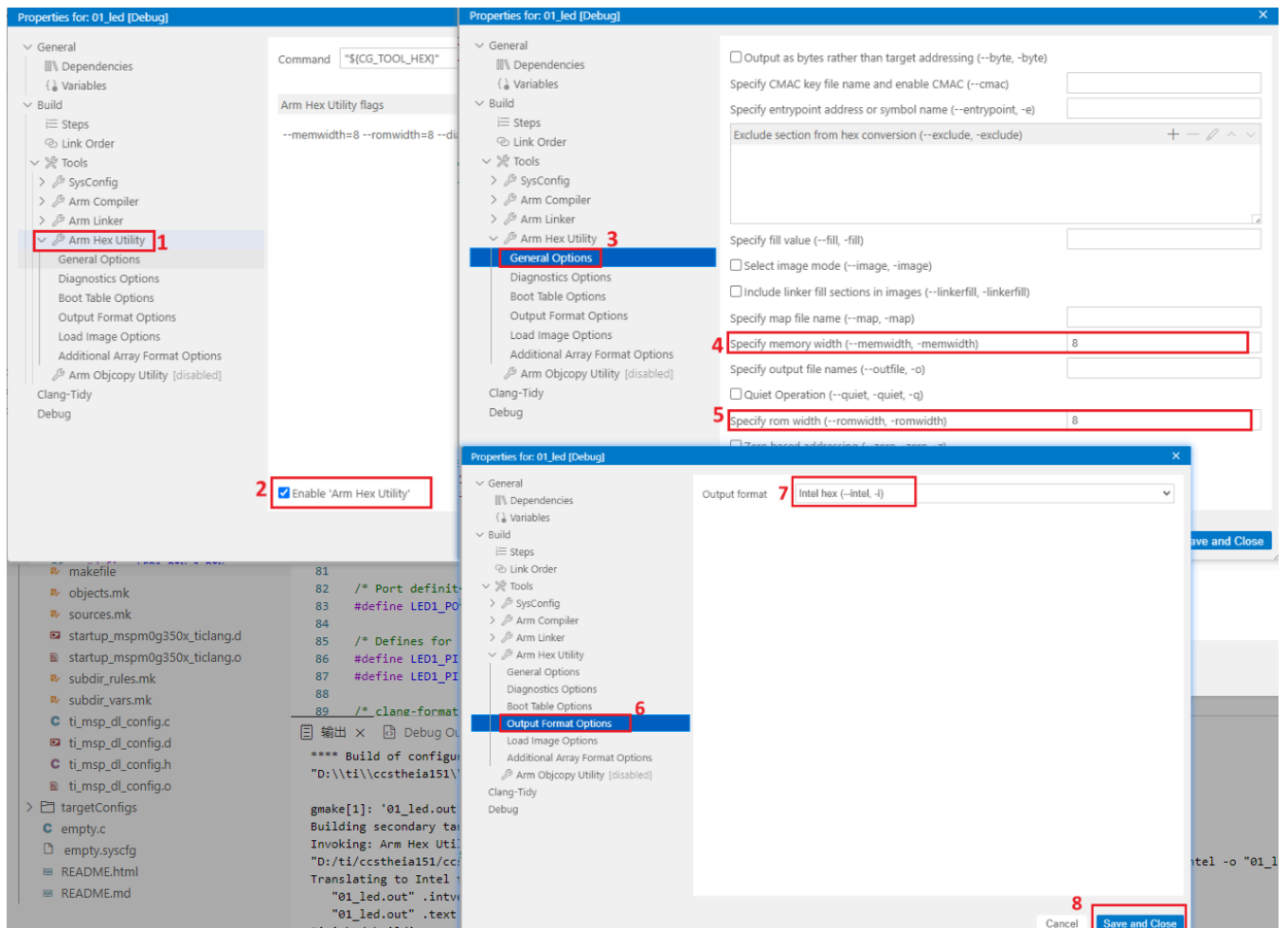
7. Build Project

7.1 Build Configuration

Right-click the project file, then select **Properties**. Select **Build -->Tools --> Arm Hex Utility**, check **Enable**.

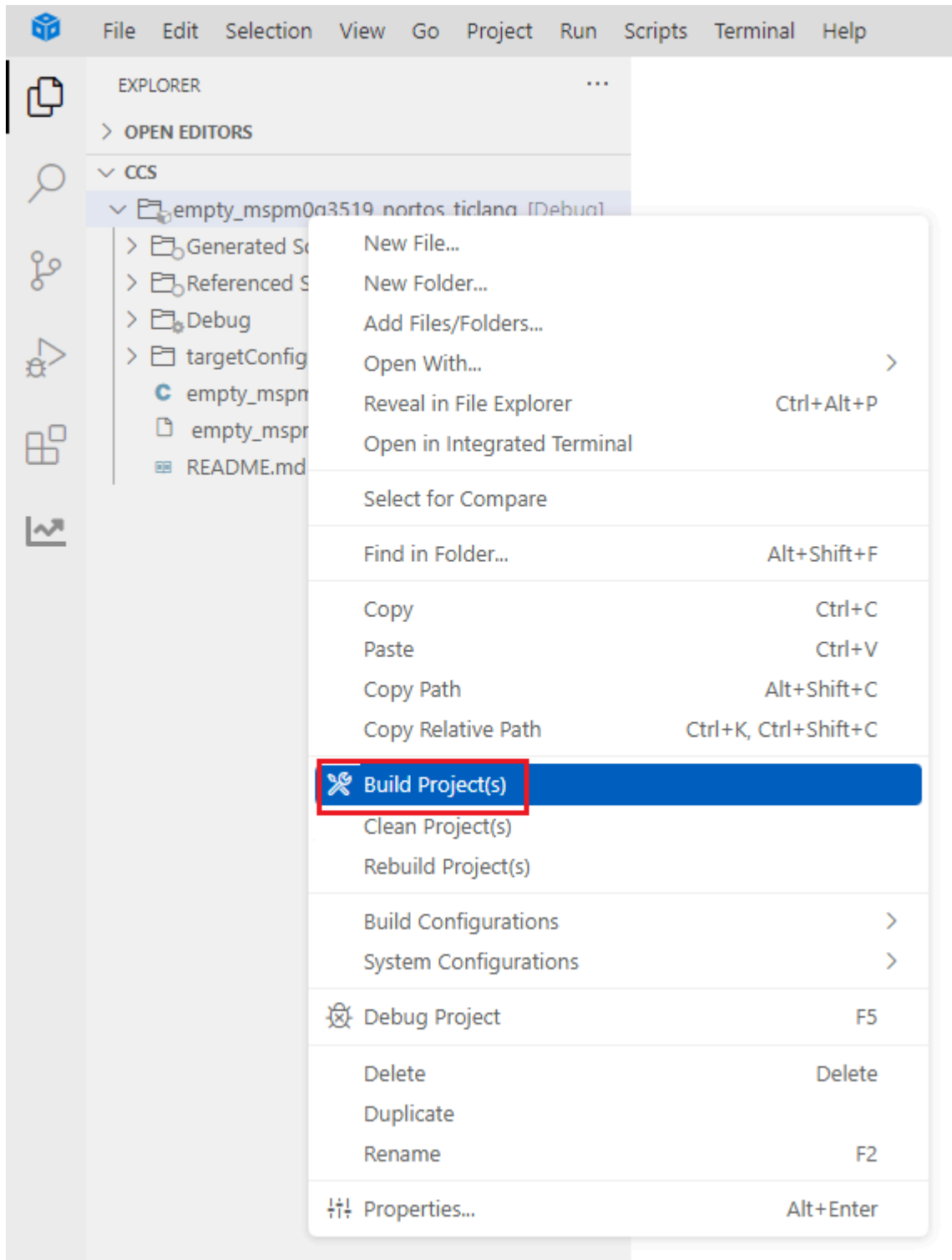
In **General Options**, select memory and ROM width of 8. In **Output Format Options**, select output format **Intel hex (--intel, -i)**. Finally, save and exit.





7.2 Build

Right-click the project folder, select **Build Project(s)** to build.



Normal build output is as follows:

```

Output x
-----
D:/ti/ccs151/ccs/tools/compiler/ti-cgt-armllvm_4.0.0.LTS/bin/tiarmclang.exe" @"syscfg/device.opt" -march=thumbv6m -mcpu=cortex-m0plus -mfloat-abi=soft -mlittle-endian -mthumb -O2 -gdwarf-3 -Wl,-m"empt
Finished building target: "empty_mspm0g3519_nortos_ticlang.out"

Building secondary target: "empty_mspm0g3519_nortos_ticlang.hex"
Invoking: Arm Hex Utility
D:/ti/ccs151/ccs/tools/compiler/ti-cgt-armllvm_4.0.0.LTS/bin/tiarmhex.exe" --memwidth=8 --romwidth=8 --diag_wrap=off --intel -o "empty_mspm0g3519_nortos_ticlang.hex" "empty_mspm0g3519_nortos_ticlang.o
Translating to Intel format...
"empty_mspm0g3519_nortos_ticlang.out".intvecs ==> .intvecs
"empty_mspm0g3519_nortos_ticlang.out".text ==> .text
Finished building secondary target: "empty_mspm0g3519_nortos_ticlang.hex"

**** Build Finished ****

```